
intelligent job scheduling for hpc systems

A statistical evaluation of deep reinforcement learning approaches. Honours 2026, Submission 1.

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This deck reflects the state of the work at Submission 1, April 2026. It is restyled, not revised: the questions it asks were open at the time and are answered in Submission 3 and the final paper.

problem context & literature gap

HPC SCHEDULING CHALLENGE



TOP500 heterogeneous systems growth.

- Traditional heuristics inflexible
- Meta-heuristics don't generalise
- DRL offers adaptive policy learning

— A field that agreed on an answer before it measured one

CRITICAL LITERATURE GAP

59

papers surveyed, 2024–2025

46%

27 papers use PPO-based approaches

20%

only 12 papers empirically justify the choice

Most cite “stability” without comparative evidence.

RESEARCH IMPLICATION

**methodological
convergence without
empirical validation
suggests algorithmic path
dependence**

PRIMARY RESEARCH QUESTION

is the field's preference for masked ppo variants empirically justified compared to other drl algorithm families?

STATISTICAL FRAMEWORK

Shapiro–Wilk	Normality check
Friedman test	Omnibus significance
Nemenyi post-hoc	Pairwise comparisons
CD diagrams + ϵ^2	Visualisation and effect size

Reference: Demšar 2006 methodology.

ALGORITHM SELECTION

ALGORITHM	FAMILY	MASKING
☒ MaskablePPO	Policy gradient	Yes
☒ MaskableDQN	Value-based	Yes
☐ Vanilla PPO	Policy gradient	No
☐ A2C	Actor–critic	No

4 algorithms × 2 real Slurm traces (84k + 28k jobs)

— Four algorithms, two traces, and a rank-based framework to judge them

current progress

STATUS

COMPLETED INFRASTRUCTURE

Gymnasium simulation environment	HPCsim with action masking support
Classical baseline evaluation	42 combinations (7 selectors × 3 allocators × 2 partitions)
Statistical analysis framework	Friedman and Nemenyi pipeline designed
Paper writing	Introduction and literature review sections complete
Visualisation pipeline	visualise.py for multi-run analysis

BASELINE RESULTS · PHYSICAL CLUSTER, 84K CPU JOBS

ALGORITHM	AVG WAIT (S)	SLOWDOWN
LCFS + best_fit	1,851	12.71
UNICEP + best_fit	1,955	12.78
SJF + best_fit	1,988	13.32
FCFS + best_fit	2,098	15.21

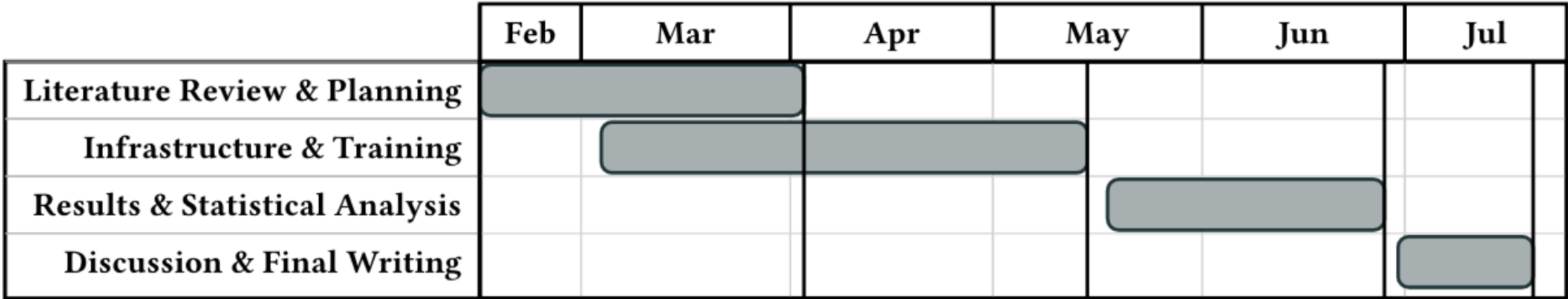
REFERENCE POINT

drl must beat 1,851 s average wait to justify its complexity

The bar the learned policies have to clear

research plan & timeline

PLAN



Submission 1
Apr 03

Submission 2
May 15

Submission 3
Jun 28

Final Submission
Jul 20

NEXT STEPS · SIX WEEKS TO SUBMISSION 2

Weeks 1–2	Setting up DRL training infrastructure
Weeks 3–5	DRL training and setting up statistical analysis (Friedman tests, CD diagrams)
Week 6	Writing Submission 2, updating presentation and website

— Six weeks, roughly 400 GPU-hours, and a training run to get right

COMPUTATIONAL RESOURCES

Training budget	~400 GPU-hours estimated
Wall-clock time	~9 days with parallelisation
GPU allocation	2-week period, to be confirmed

expected contribution ■

CONTRIBUTION

PRIMARY CONTRIBUTION

first rigorous statistical comparison of drl algorithm families for hpc scheduling on real heterogeneous workloads

IF PPO WINS

Empirical validation for the field's convergence.

Broader impact: a foundation for distillation and deployment in resource-constrained environments.

IF AN ALTERNATIVE WINS

Challenge conventional assumptions.

— Either answer is worth having, which is why the comparison is worth running

references

SOURCES CITED IN THIS SUBMISSION

SYSTEMS & WORKLOADS

TOP500 (2026). TOP500 List Statistics. top500.org/statistics/list. Accelerator share, slide 2.

Yoo, Jette & Grondona (2003). SLURM: Simple Linux Utility for Resource Management. JSSPP, LNCS 2862, 44–60.

STATISTICS

Demšar (2006). Statistical Comparisons of Classifiers over Multiple Data Sets. JMLR 7, 1–30. The framework proposed here.